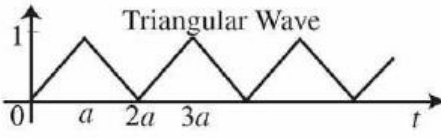
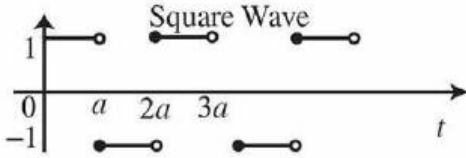
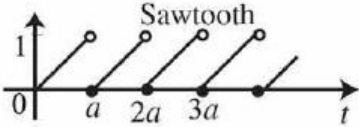


Table of Laplace Transforms

Notice: We use $L(s)$ instead of $F(s)$ in our Lecture slides.

$f(t), \quad t \geq 0$	$F(s) = \mathcal{L}(f)(s) = \int_0^\infty f(t)e^{-st} dt,$
1. 1	$\frac{1}{s}, \quad s > 0$
2. $t^n, \quad n = 1, 2, \dots$	$\frac{n!}{s^{n+1}}, \quad s > 0$
3. $t^a \quad (a > -1)$	$\frac{\Gamma(a+1)}{s^{a+1}}, \quad s > 0$
4. e^{at}	$\frac{1}{s-a}, \quad s > a$
5. $t^n e^{at}$	$\frac{n!}{(s-a)^{n+1}}, \quad s > a$
6. $\frac{e^{at} - e^{bt}}{a-b}$	$\frac{1}{(s-a)(s-b)}, \quad s > \max(a, b)$
7. $\frac{ae^{at} - be^{bt}}{a-b}$	$\frac{s}{(s-a)(s-b)}, \quad s > \max(a, b)$
8. $\sin kt$	$\frac{k}{s^2 + k^2}, \quad s > 0$
9. $\cos kt$	$\frac{s}{s^2 + k^2}, \quad s > 0$
10. $e^{at} \sin kt$	$\frac{k}{(s-a)^2 + k^2}, \quad s > a$
11. $e^{at} \cos kt$	$\frac{s-a}{(s-a)^2 + k^2}, \quad s > a$
12. $t \sin kt$	$\frac{2ks}{(s^2 + k^2)^2}, \quad s > 0$
13. $t \cos kt$	$\frac{s^2 - k^2}{(s^2 + k^2)^2}, \quad s > 0$
14. $\frac{1}{2a^3}(\sin at - at \cos at)$	$\frac{1}{(s^2 + a^2)^2}, \quad s > 0$
15. $\sinh kt$	$\frac{k}{s^2 - k^2}, \quad s > k $
16. $\cosh kt$	$\frac{s}{s^2 - k^2}, \quad s > k $
17. $e^{at} \sinh kt$	$\frac{k}{(s-a)^2 - k^2} \quad s > a + k $
18. $e^{at} \cosh kt$	$\frac{s-a}{(s-a)^2 - k^2} \quad s > a + k $
19. $t \sinh kt$	$\frac{2ks}{(s^2 - k^2)^2}, \quad s > k $
20. $t \cosh kt$	$\frac{s^2 + k^2}{(s^2 - k^2)^2}, \quad s > k $
21. $\frac{1}{2k^3}(kt \cosh kt - \sinh kt)$	$\frac{1}{(s^2 - k^2)^2}, \quad s > k $

$f(t), \quad t \geq 0$	$F(s) = \mathcal{L}(f)(s) = \int_0^\infty f(t)e^{-st} dt,$
22. $\delta_0(t - t_0), \quad t_0 \geq 0$	$e^{-t_0 s}, \quad s > 0$
23. $\mathcal{U}_0(t - a) = \begin{cases} 0 & \text{if } t < a \\ 1 & \text{if } t \geq a \end{cases} \quad (a > 0)$	$\frac{e^{-as}}{s}, \quad s > 0$
24. $f(t + T) = f(t) \quad (T > 0)$	$\frac{1}{1 - e^{-Ts}} \int_0^T e^{-st} f(t) dt$
25. $f(t + T) = -f(t) \quad (T > 0)$	$\frac{1}{1 + e^{-Ts}} \int_0^T e^{-st} f(t) dt$
26. 	$\frac{1}{as^2} \left[\frac{1 - e^{-as}}{1 + e^{-as}} \right] = \frac{1}{as^2} \tanh\left(\frac{as}{2}\right), \quad s > 0$
27. 	$\frac{1}{s} \left[\frac{1 - e^{-as}}{1 + e^{-as}} \right] = \frac{1}{s} \tanh\left(\frac{as}{2}\right), \quad s > 0$
28. 	$\frac{1}{as^2} - \frac{e^{-as}}{s(1 - e^{-as})}, \quad s > 0$
29. $\frac{\sin at}{t}$	$\tan^{-1}\left(\frac{a}{s}\right), \quad s > 0$
30. $J_0(at)$	$\frac{1}{\sqrt{s^2 + a^2}}, \quad s > 0$
31. $J_0(a\sqrt{t})$	$\frac{e^{-a^2/4s}}{s}, \quad s > 0$
32. $t^p J_p(at) \quad (p > -\frac{1}{2})$	$\frac{2^p a^p \Gamma(p + \frac{1}{2})}{\sqrt{\pi}(s^2 + a^2)^{p + \frac{1}{2}}}, \quad s > 0$
33. $\frac{\sqrt{\pi}}{\Gamma(k)} \left(\frac{t}{2a}\right)^{k - \frac{1}{2}} J_{k - \frac{1}{2}}(at) \quad (k > 0)$	$\frac{1}{(s^2 + a^2)^k}, \quad s > 0$
34. $\frac{\sqrt{\pi}}{\Gamma(k)} a \left(\frac{t}{2a}\right)^{k - \frac{1}{2}} J_{k - \frac{3}{2}}(at) \quad (k > \frac{1}{2})$	$\frac{s}{(s^2 + a^2)^k}, \quad s > 0$
35. $2 \sum_{m=1}^n \binom{2n - m - 1}{n - 1} \frac{t^{m-1} \cos(at - \frac{m\pi}{2})}{(2a)^{2n-m}(m-1)!}$ (n an integer ≥ 1)	$\frac{1}{(s^2 + a^2)^n}, \quad s > 0$
36. $\frac{1}{(n-1)!} \sum_{m=1}^{n-1} \frac{(2n - m - 3)!}{(m-1)!(n-m-1)!} \frac{t^m \cos(at - \frac{m\pi}{2})}{(2a)^{2n-m-2}}$ (n an integer ≥ 2)	$\frac{s}{(s^2 + a^2)^n}, \quad s > 0$

$f(t), \quad t \geq 0$	$F(s) = \mathcal{L}(f)(s) = \int_0^\infty f(t)e^{-st} dt,$
37. $\operatorname{erf}(at) \quad (a > 0)$	$\frac{1}{s} e^{s^2/4a^2} \operatorname{erfc}\left(\frac{s}{2a}\right), \quad s > 0$
38. $\operatorname{erf}(a\sqrt{t})$	$\frac{a}{s\sqrt{s+a^2}}, \quad s > 0$
39. $e^{-a^2 t^2} \quad (a > 0)$	$\frac{\sqrt{\pi}}{2a} e^{s^2/4a^2} \operatorname{erfc}\left(\frac{s}{2a}\right), \quad s > 0$
40. $\frac{1}{\sqrt{\pi t}} e^{-a^2/4t} \quad (a \geq 0)$	$\frac{e^{-a\sqrt{s}}}{\sqrt{s}}, \quad s > 0$
41. $\frac{a}{2\sqrt{\pi t^{3/2}}} e^{-a^2/4t} \quad (a > 0)$	$e^{-a\sqrt{s}}, \quad s > 0$
42. $\operatorname{erfc}\left(\frac{a}{2\sqrt{t}}\right) \quad (a \geq 0)$	$\frac{1}{s} e^{-a\sqrt{s}}, \quad s > 0$
Operational Properties	
43. $\alpha f(t) + \beta g(t)$	$\alpha F(s) + \beta G(s)$
44. $f'(t)$	$sF(s) - f(0)$
45. $f''(t)$	$s^2 F(s) - sf(0) - f'(0)$
46. $f^{(n)}(t)$	$s^n F(s) - s^{n-1}f(0) - \dots - f^{(n-1)}(0)$
47. $-tf(t)$	$F'(s)$
48. $t^n f(t)$	$(-1)^n F^{(n)}(s)$
49. $\int_0^t f(\tau) d\tau$	$\frac{1}{s} F(s), \quad s > 0$
50. $\int_0^t \int_0^\tau f(\rho) d\rho d\tau$	$\frac{1}{s^2} F(s), \quad s > 0$
51. $\frac{f(t)}{t}$	$\int_s^\infty F(u) du$
52. $\frac{f(t)}{t^2}$	$\int_s^\infty \int_\sigma^\infty F(u) du d\sigma$
53. $\mathcal{U}_0(t-a)f(t-a) \quad (a > 0)$	$e^{-as} F(s)$
54. $e^{at} f(t)$	$F(s-a)$
55. $f(ct) \quad (c > 0)$	$\frac{1}{c} F\left(\frac{s}{c}\right)$
56. $f * g(t) = \int_0^t f(\tau)g(t-\tau) d\tau$	$F(s)G(s)$

Source:

Nakhlé H. Asmar, Appendix B, *Partial Differential Equations with Fourier Series and Boundary Value Problems, 2nd Edition, 2004*